



Magic Touch

DENTAL LAB SOLUTIONS

DLCPM Enterprise



Production Management User Guide

Introduction:

Production Management is a tool used by technicians and managers to, at a glance, see what the current state of production is in your lab. In order to complete a product, the lab performs a set of tasks that technicians complete. In Production Management a group of tasks will be considered a Production Stage. Products, tasks, and stages are all dynamically defined in DLCPM Enterprise.

For example, a product within the chain of production will have 12 tasks split into 4 distinct stages.

Stage 1			Stage 2			Stage 3			Stage 4		
Task 10	Task 20	Task 30	Task 40	Task 50	Task 60	Task 70	Task 80	Task 90	Task 100	Task 110	Task 120

As technicians complete a product's tasks in DLCPM Enterprise, the Production Management tool will move that product through each stage.

Requirements:

To use the Production Management Utility, labs must be subscribed to DLCPM Enterprise. Both hosted and on-Premises labs have access to Production Management.

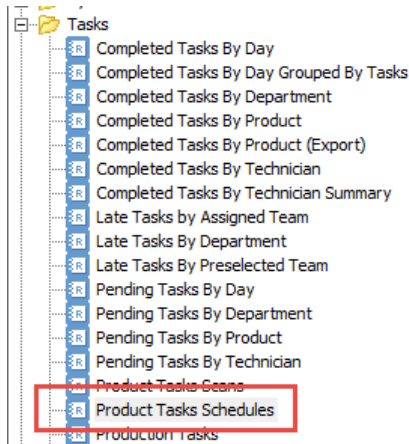
Desktop Computers
- Windows 10 or 11 Operating System
- Computers with i5, i7 CPU equivalent or better processor
- 8GB RAM or better
- 250GB or more HDD or SSD

Planning

Some preliminary planning may be required to utilize Production Management to its full capacity. Each product schedule will need to be separated into stages. This can be done prior to defining the lab's Production Stages in DLCPM Enterprise.

To aid in setting up Production Stages, the "Product Tasks Schedules" report is available to generate a complete list of the current production schedules for all products within DLCPM Enterprise. This will give you an overview of all tasks set for each product so you can begin planning out the lab's stages for each product schedule.

The report can be run by navigating to the Report Center in DLCPM Enterprise and under the 'Tasks' folder.



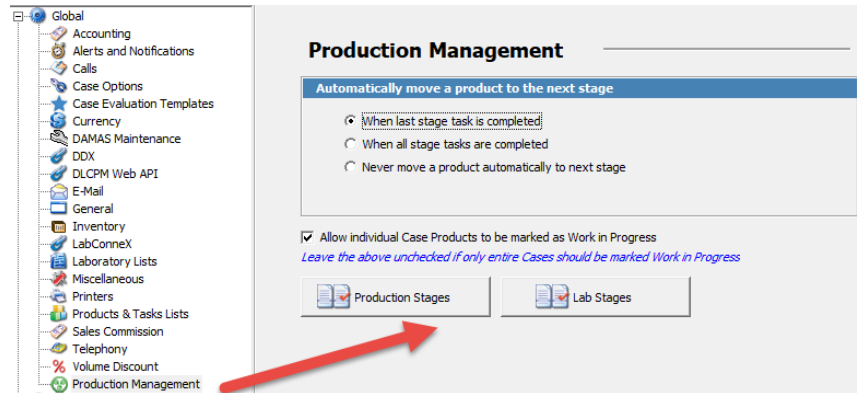
If no schedules exist in your DLCPM Enterprise environment and you plan on using the Production Management tool, the lab should consider setting these stages as you build your production schedules.

The Production Management tool can be used with Load Schedule Assignment, but it is not required. This tool will still keep track of case stages if tasks are assigned manually.

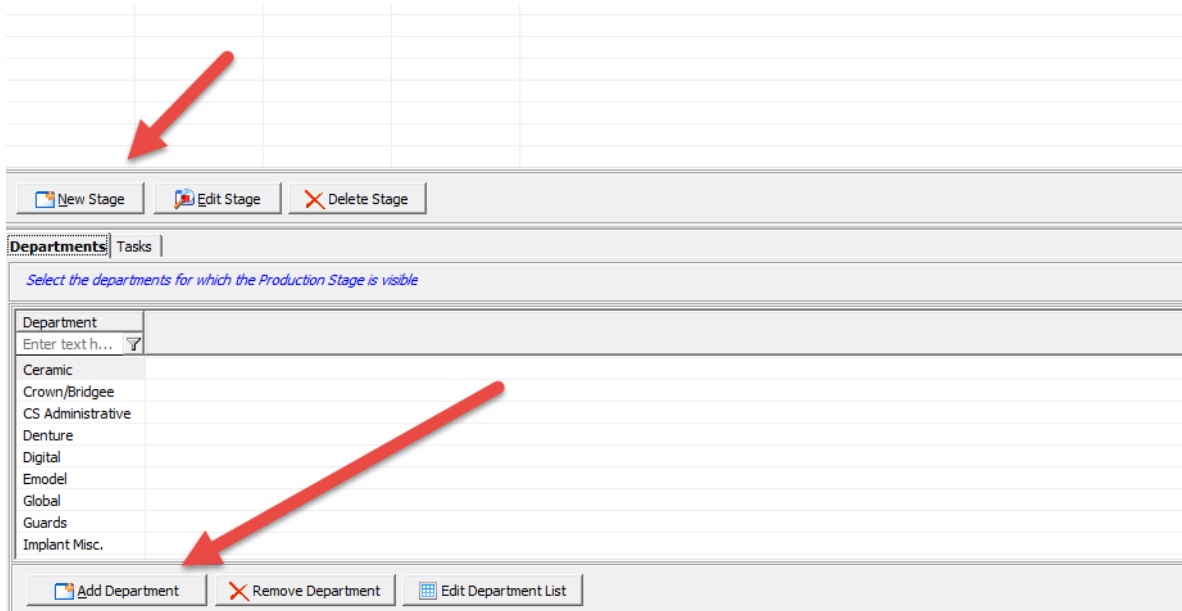
If the lab does not use Load Schedule Assignment, Production Management can still help facilitate all assignments to technicians by the production management team at the lab.

Defining the Lab's Production Stages

The first step in setting up the Production Management will be to define the lab's Production Stages. These can be defined in Global Settings > Production Management > Production/Lab Stages.



In the Production Stages menu, labs can add new stages and set their departments. Lab Stages can also be set up if there are multiple lab entities that managers need to track.



Each production schedule may have different departments and can be set to display accordingly. At this point, you can choose to color code each stage in DLCPM Enterprise.

Please keep in mind that Product Tasks set with production stages will override stages set in Production Tasks.

Associating Stages with specific Colors

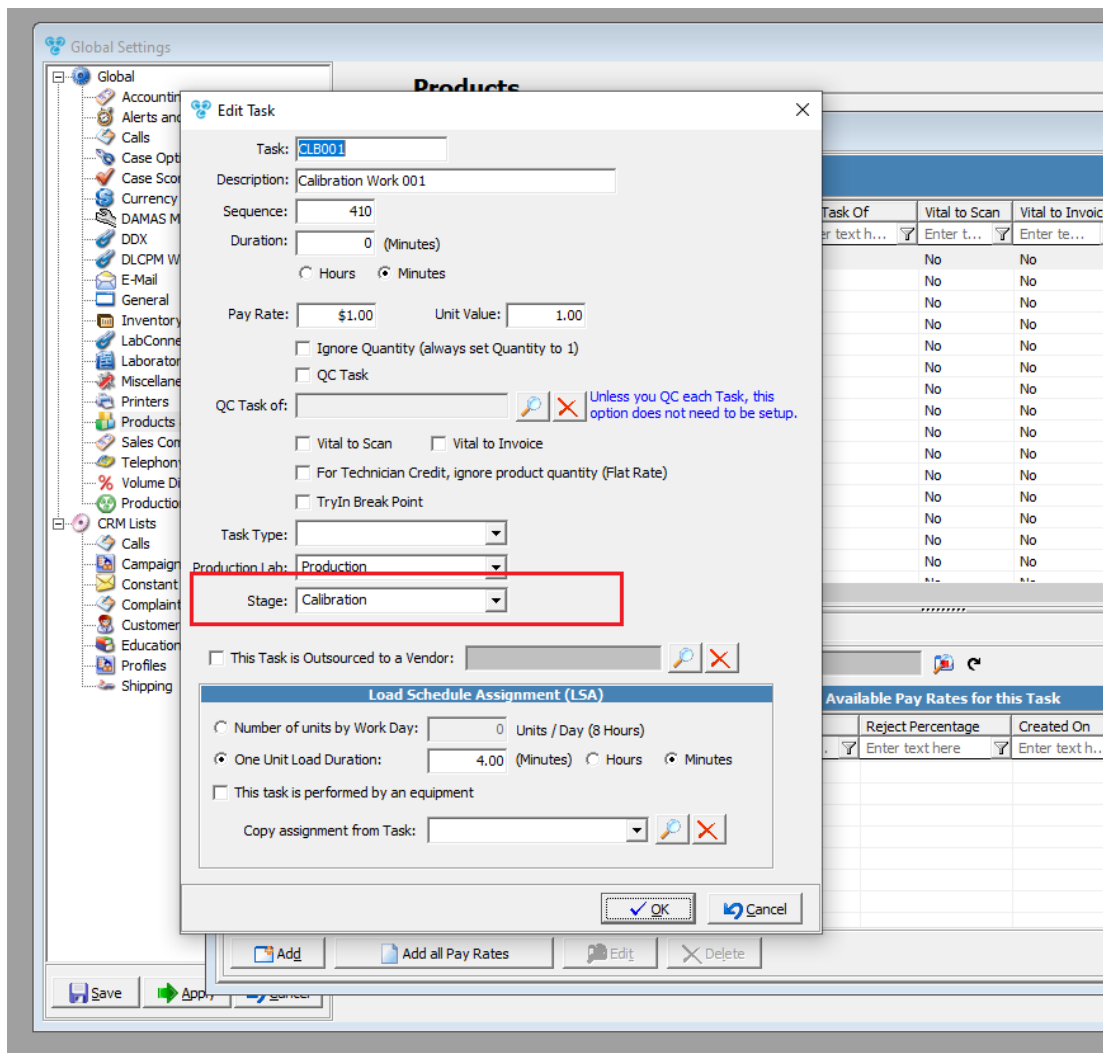
Users can associate a specific color with each stage. This makes it easier to identify the stages. Whenever a stage is displayed in a grid, it will be dynamically displayed using its associated color. If a stage has no color associated, then it will be displayed as normal text (black text on white background).

Whenever a stage is selected in the tree view, a title with the stage name will be displayed on the task grid and the appropriate colors will be used. We have a fixed number of colors from which users can pick when they add or edit a Stage. Users can choose to use the same color for two different stages.

Color Label	Sample
Amber	Text
Brown	Text
Dark Blue	Text
Dark Green	Text
Gold	Text
Gray	Text
Green	Text
Light Blue	Text
Light Green	Text
Orange	Text
Pink	Text
Red	Text
Silver	Text
Violet	Text
Yellow	Text

Case Number	Product ID	Due Date	Ship Date	Stage	
64195	PM4	3/16/2023	3/16/2023	1	
64196	PM1	3/17/2023	3/17/2023	2	
64194	PM2	3/16/2023	3/16/2023	3	
64197	PM1	3/17/2023	3/17/2023	4	
64198	PM1	3/17/2023	3/17/2023	5	

Once the stage lists are defined, you can begin to set these stages in your Production Tasks.



To set these stages, navigate to Global Settings > Products > Production Tasks. If you plan on using this as the lab's Global Stage setting then you can choose to only set them on the Production Task level.

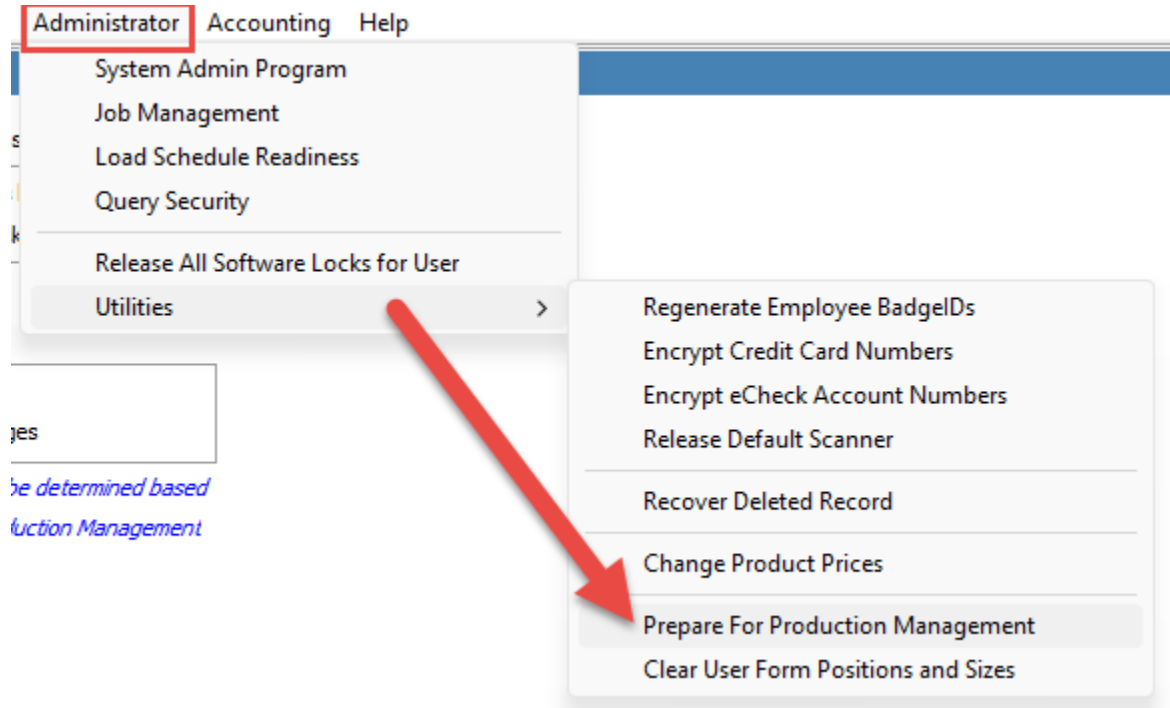
Stages can be set up with your current production schedules or new stages defined as you move through the process of setting them up.

If the tasks' stages are going to be different for each product schedule, these stages can be set up per product. Navigate to the product you wish to update and set the stages in the Product Tasks tab > Edit Product Task menu. If the Production Stages are going to be the same for all related products, you can skip this step.

Preparing DLCPM Enterprise for Production Management

The final step in this process will be to prepare the DLCPM Enterprise database for the use of the Production Management tool. There are a few options to complete this.

Navigate to Administrator > Utilities > Prepare for Production Management.



There are two options available.

Execute the following actions

☒ Set the Stage on Products Schedule (based on Task Stage)

☒ Only where Product Task Stage is blank

☐ Overwrite all existing Product Task Stages

☒ Set the Stage on Open Cases

☒ Only where Stage is blank

☐ Overwrite all existing current Stages

The current Case Product Stage will be determined based on Method and Global Settings - Production Management

These options will populate all product schedules/cases within DLCPM Enterprise with the stages you have chosen for each task/schedule.

This first option below will only need to be run if you are using Global Production Tasks to populate Product Schedules.

1. "Set the Stage on Product's Schedule (based on Task Stage)"
 - a. Only where Product Task Stage is blank - This will only populate product schedules where the stage is blank. This is the safer method.

Execute the following actions

☒ Set the Stage on Products Schedule (based on Task Stage)

- ☒ Only where Product Task Stage is blank

☐ Overwrite all existing Product Task Stages

- b. The second option "Overwrite all existing production stages" will wipe all data set in individual products and set them to what is defined at the Production Task level.

Please note that this action cannot be reversed and doing so will wipe all configurations set in the Product Schedules/Tasks.

2. "Set the Stage on Open Cases"
 - a. Only where Stage is blank - This will find all cases that have blank values set for stages and update them, so they are displayed in Production Management.

☒ Set the Stage on Open Cases

- ☒ Only where Stage is blank

☐ Overwrite all existing current Stages

The current Case Product Stage will be determined based on Method and Global Settings - Production Management

- b. Overwrite all existing current Stages - This option will wipe all the current data set for cases within DLCPM Enterprise.

Please note that this action cannot be reversed and doing so will wipe all configurations set in cases.

Running Production Management

From DLCPM Enterprise, users can access the Tools menu and select Production Management or use the icon in the left panel under the Main tab.



Production Management is also a stand-alone application that can be started from a desktop icon.

The main screen consists of:

- Toolbar (to search, filter, export records, configure the program, and do different actions on products)
- 2 top tabs: Summary, Production Management
- “Work in Progress” section at the bottom.

Production Management 1.144 Harout@Harout

Stages: Dashboard

Lab: -ALL- Product: -ALL- Show Cases Out of Scope Ship Date: From: 08-30-2024 To: 09-13-2024

Case # Stage Department: -ALL Department- Show Cases On Hold Due Date: Refresh Apply Filters Jump To Stage Complete Task Write Notes Grid Filters Save Grid Filters User Configuration

Summary Production Management

Drag a column header here to group by that column

Case Number	Product ID	Production Lab	Stage	Pan Number	Due Date	Ship Date	Status	Customer ID	Quantity	Description	Teeth Numbers	General Instruct...	On Hold	out Of Scope	Rush	Priority
66589	1001	MagicTouch	CB Product Rx De...		9/5/2024	8/28/2024	In Production	1000	1	Argeloy NP Special	2	PD Level Pref	☐	☐	☐	☐
66590	1001	MagicTouch	CB Product Rx De...		9/5/2024	8/28/2024	In Production	1000	1	Argeloy NP Special	3	PD Level Pref	☐	☐	☐	☐
63610	1001	MagicTouch	CB Hold		9/2/2024	8/30/2024	In Production	1078	0	Argeloy NP Special			☐	☐	☐	☐
66600	1001	MagicTouch	CB SG Hold for Pa...		9/2/2024	8/30/2024	In Production	1049	1	Argeloy NP Special	5		☐	☐	☐	☐
66597	1001	MagicTouch	CB Product Rx De...		9/2/2024	8/30/2024	In Production	1009	1	Argeloy NP Special	5		☐	☐	☐	☐
66596	1001	MagicTouch	CB Product Rx De...		9/2/2024	8/30/2024	In Production	1008	1	Argeloy NP Special	3		☐	☐	☐	☐
66598	1001	MagicTouch	CB Hold		9/3/2024	8/30/2024	In Production	1016	1	Argeloy NP Special	3		☐	☐	☐	☐
66603	1001	MagicTouch	CB Product Rx De...		9/2/2024	8/30/2024	In Production	1080	1	Argeloy NP Special	3		☐	☐	☐	☐
66599	1001	MagicTouch	CB Surgical Guide ...		9/3/2024	8/30/2024	In Production	1016	1	Argeloy NP Special	5		☐	☐	☐	☐
66601	1001	MagicTouch	CB Hold		9/2/2024	8/30/2024	In Production	1050	1	Argeloy NP Special	5		☐	☐	☐	☐

Work In Progress

WIP Employee Full Name WIP Employee ID

Case Number	Product ID	Production Lab	Stage	Pan Number	Due Date	Ship Date	Status	Customer ID	Quantity	Description	Teeth Numbers	General Instruct...	On Hold	out Of Scope	Technician Name	Rush	Priority
Dianne B Constant; 12507 Products :1, Quantity :1																	
66595	1001	MagicTouch	CB Product Rx ...		9/2/2024	8/30/2024	In Production	1008	1	Argeloy NP Spe...	2		☐	☐		☐	☐
Manuel Margot; 07 Products :2, Quantity :2																	
66602	1001	MagicTouch	CB SG Send to P...		9/2/2024	8/30/2024	In Production	1050	1	Argeloy NP Spe...	6		☐	☐		☐	☐
63609	1001	MagicTouch	CB Surgical Guid...		9/2/2024	8/30/2024	In Production	1078	1	Argeloy NP Spe...	14		☐	☐		☐	☐

Production Management Toolbar –

The main menu toolbar gives users the ability to filter out cases/products based on the selection made in the filters. Users can choose Labs, Products, Date Ranges, and other options to refine the view to their preference.

Summary Tab –

This will be the main menu that will display all the cases/products that are currently in production in the lab. Lab users can drag and drop any column in the list to the top header and create a grouping of filters that can be applied or saved based on the user's preference.

Grid filters can be saved by creating a group of filters and hitting the "Save Grid Filters" button on the toolbar, below the "Grid Filter" drop-down. The "Grid Filter" drop-down will be blank until users save their first filter set.

Production Management Tab –

This tab allows the user to see the same information as the Summary tab, but it will break down by the task stages rather than the Product or Case. Users can see what tasks are in what stage by clicking on the respective task stage in the left-hand menu. Column Grid Filters can be applied the same way as in the Summary tab.

Work In Progress –

Production Managers can use this option to assign tasks on the fly. Simply select an entry from the top section, drag it to the Work In Progress area, and drop it. Production Management would then ask which technician would complete the work. Once the work is done, the technician can complete the tasks and the entry in Production Management can be completed by dragging and dropping that case back into the Summary area. You can choose to complete the stage, if necessary, at this point or keep it in its current stage if it hasn't been completed.

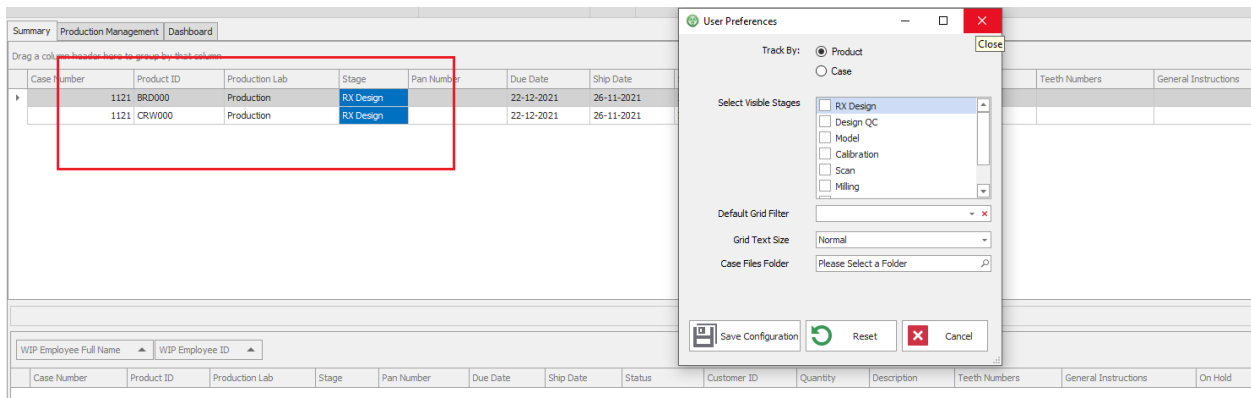
User Specific Configurations

There are a few user options that can be configured directly from Production Management. All these options are related to how the records are displayed here. Defining tasks and stages can only be done in DLCPM Enterprise and can't be changed here. Any change made here by a user will only affect that user and will not affect the other users.

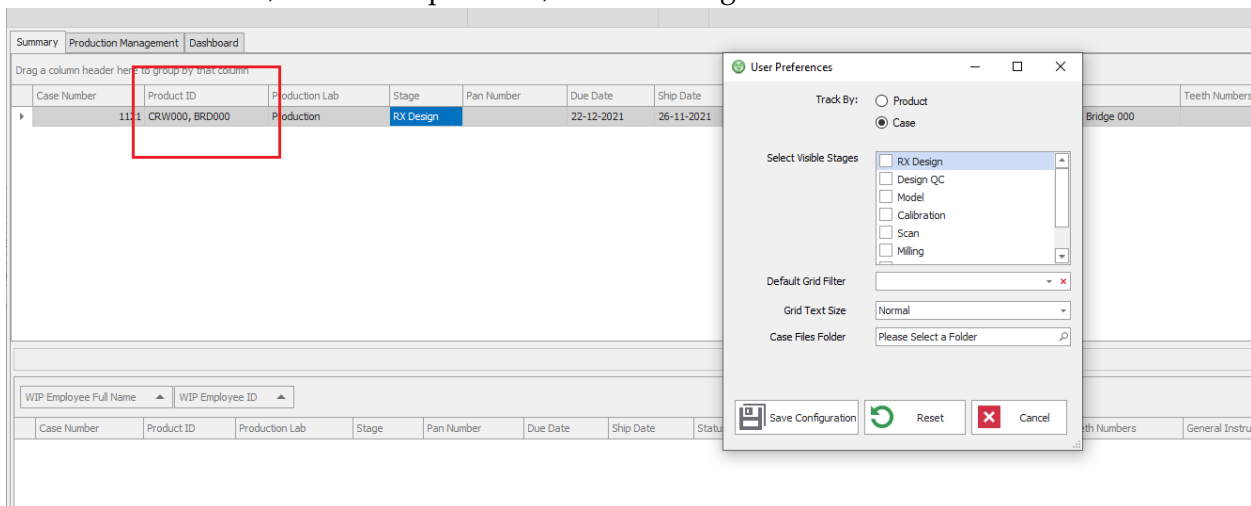
Tracking by Product or Case –

Users can choose to track individual products or to track an entire case.

If tracking is done by product, then you can have different products from the same case in different stages and they will be displayed as independent records.

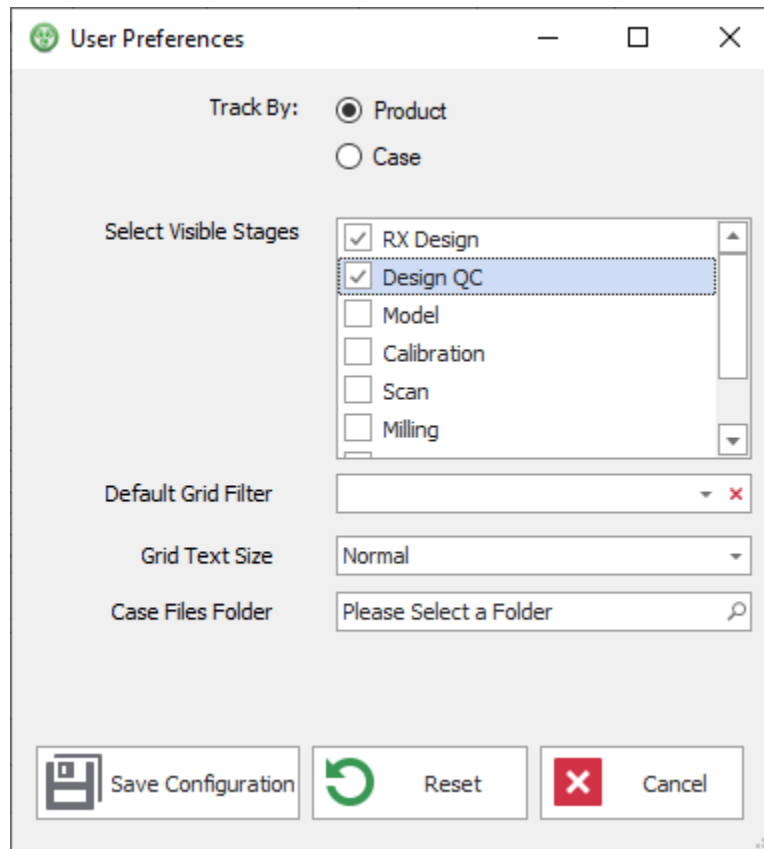


If a user is choosing tracking by Case, then all the products from the same case will be displayed as a single record. This can also affect how operations are performed because now you can only move an entire Case, with all its products, from one stage to the next.



A user can choose to see only a few specific stages that are of interest to them. For this, they will have to check those stages on Select Visible Stages. If there is no stage selected here, then all the stages are visible by default.

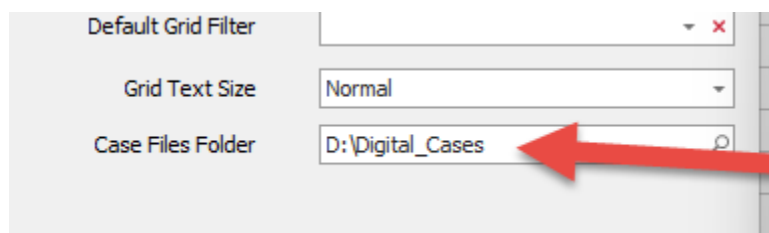
Let's say that a user chooses to only see two stages: RX Design and Design QC.



When the user reloads Production Management, they will now only see records at those specific stages.

Default Grid Filters and Grid Text Size can be adjusted in this menu as well. Make the appropriate changes and click "Save Configuration" to have these changes display the next time Production Management loads.

The Case Files Folder can be defined here as well. Click on the magnifying glass in the option and choose where you would like files to be downloaded when using Production Management.



Searching for specific cases

If you need to locate a specific case, you can type the Case Number in the Case Number Search field and hit Enter. Data will be filtered on all the pages and only that specific case will be displayed in the tool.

To view Out of Scope or On Hold cases, check the respective checkboxes and refresh the window.

Production Management 1.1.44 Harout@Harout

Stages Dashboard

Lab: -ALL- Product: -ALL- Show Cases Out of Scope Ship Date From: 08

Case # Stage Dept: -ALL Departm... Show Cases On Hold Due Date To: 09

Summary Production Management

Drag a column header here to group by that column

	Case Number	Product ID	Production ...	Stage	Pan Number	Due Date	Ship Date	Status	Customer ID	Que
▶	66589	1001	MagicTouch	CB Product ...		9/5/2024	8/28/2024	In Production	1000	
	66590	1001	MagicTouch	CB Product ...		9/5/2024	8/28/2024	In Production	1000	
	63610	1001	MagicTouch	CB Hold		9/2/2024	8/30/2024	In Production	1078	
	66600	1001	MagicTouch	CB SG Hold ...		9/2/2024	8/30/2024	In Production	1049	
	66597	1001	MagicTouch	CB Product ...		9/2/2024	8/30/2024	In Production	1009	
	66596	1001	MagicTouch	CB Product ...		9/2/2024	8/30/2024	In Production	1008	
	66598	1001	MagicTouch	CB Hold		9/3/2024	8/30/2024	In Production	1016	

Production Management 1.1.44 Harout@Harout

Stages Dashboard

Lab: -ALL- Product: -ALL- Show Cases Out of Scope

Case # 66589 Stage Dept: -ALL Departm... Show Cases On Hold

Summary Production Management

Drag a column header here to group by that column

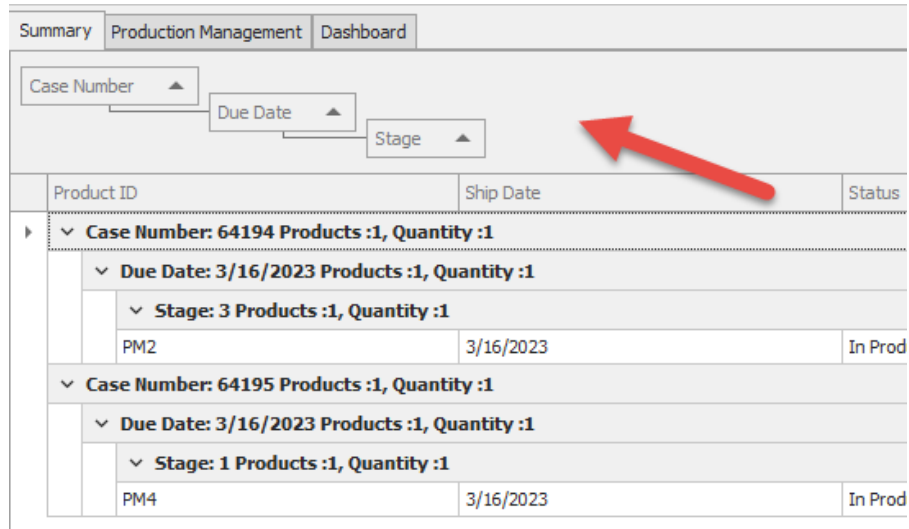
	Case Number	Product ID	Production ...	Stage	Pan Number	Due Date	Shi
▶	66589	1001	MagicTouch	CB Product ...		9/5/2024	8/2

Use the Product or Stage Dept filters to only show a certain product or stage department.

Using the Group by Feature

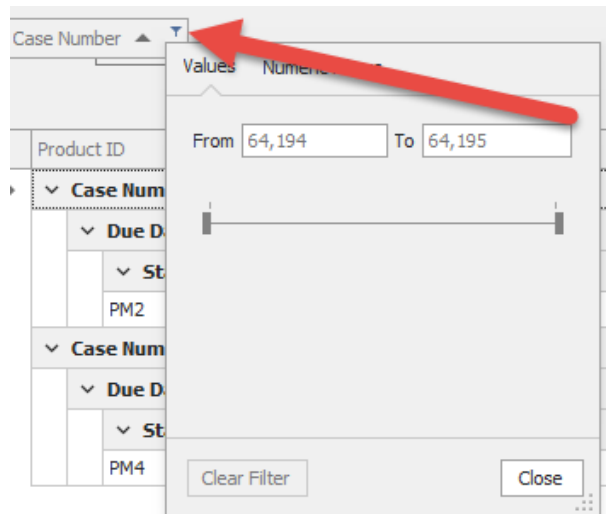
The Summary and Production Management tabs allow users to customize how the columns are grouped. Additionally, filters can be set to see only the cases that fit into the user's criteria.

If you need to see cases grouped by Case Number, Due Date, and Stage, drag and drop those columns into the top menu bar.



Summary	Production Management	Dashboard
Case Number	Due Date	Stage
Product ID	Ship Date	Status
▼ Case Number: 64194 Products :1, Quantity :1		
▼ Due Date: 3/16/2023 Products :1, Quantity :1		
▼ Stage: 3 Products :1, Quantity :1		
PM2	3/16/2023	In Prod
▼ Case Number: 64195 Products :1, Quantity :1		
▼ Due Date: 3/16/2023 Products :1, Quantity :1		
▼ Stage: 1 Products :1, Quantity :1		
PM4	3/16/2023	In Prod

To filter these results, click the filter icon at the top right of the column boxes and choose the filters you would like to apply.



Case Number

Values

From 64,194 To 64,195

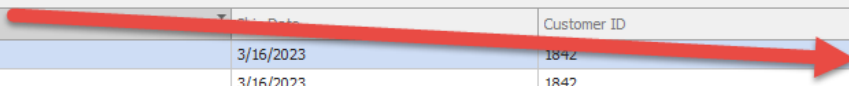
Clear Filter

Close

Managing Columns

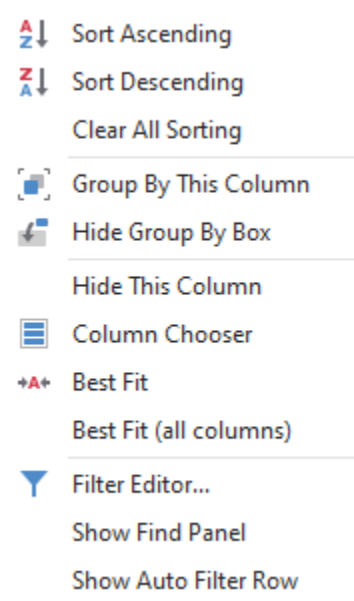
To rearrange columns in Production Management, users can drag and drop columns to the positions they like.

Drag and drop columns into the desired position:

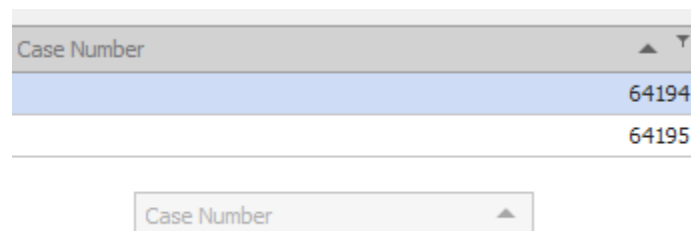


Product ID	Order Date	Customer ID	Quantity
PM2	3/16/2023	1842	Product ID
PM4	3/16/2023	1842	

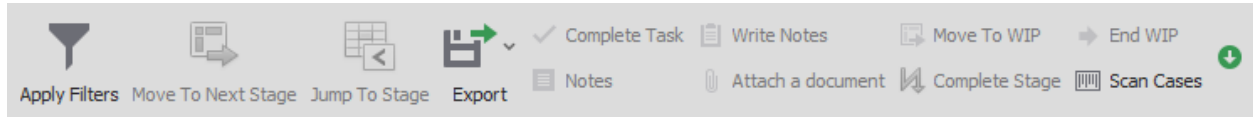
Right-clicking on the column will display the column options. There are a variety of options in the menu that will help design the Production Management tool to your preference.



To remove columns, click and drag them out of the column into the grid.

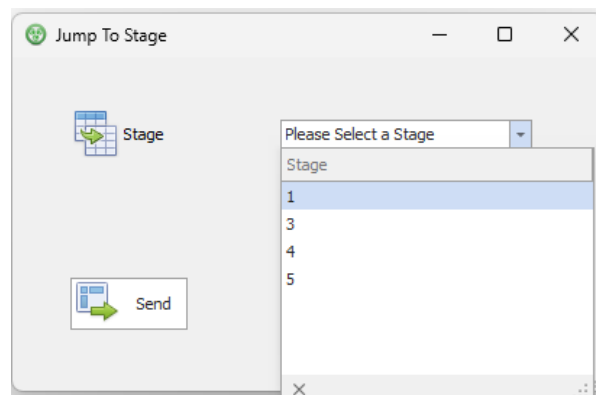


Performing operations on the toolbar

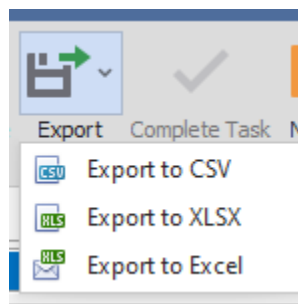


Users can select one or more products and perform different operations on those records using the Production Management tool. Some options are only available in the Production Management tab with a case selected in the grid. Some of these options are also available by right-clicking on cases in the grid.

1. Apply Filters - This feature will apply any filters chosen.
2. Move to Next Stage - Products will be moved to the next stage according to the task sequence and stages defined.
3. Jump to Stage - This option allows users to move the case/product to any stage in the sequence.

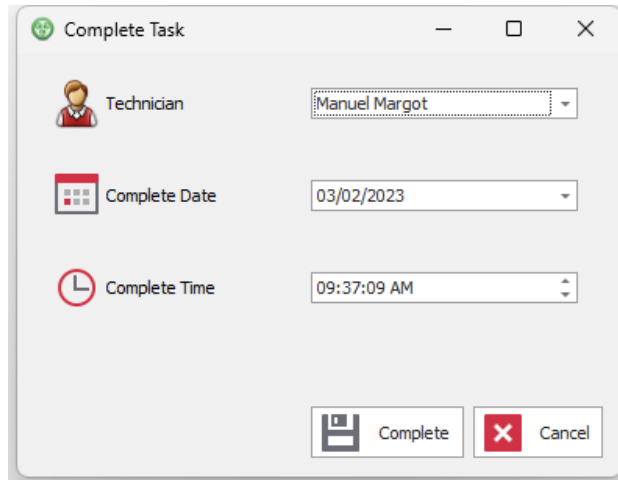


4. Export - Export all cases in the grid to a spreadsheet (CSV, XLSX, or Excel)



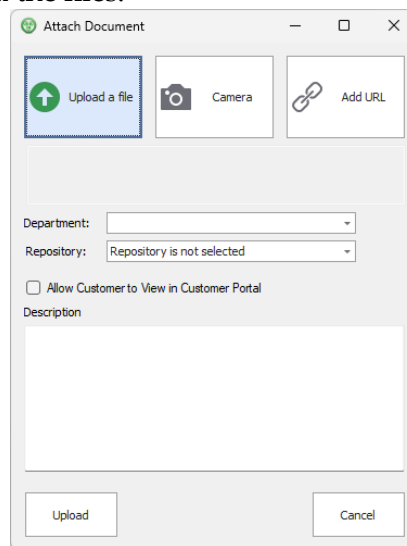
5. Complete Task

- a. Allows users to complete the next task in the stage.
- b. Confirm Technician, Complete Date and Complete Time, then click Complete to complete specified tasks.



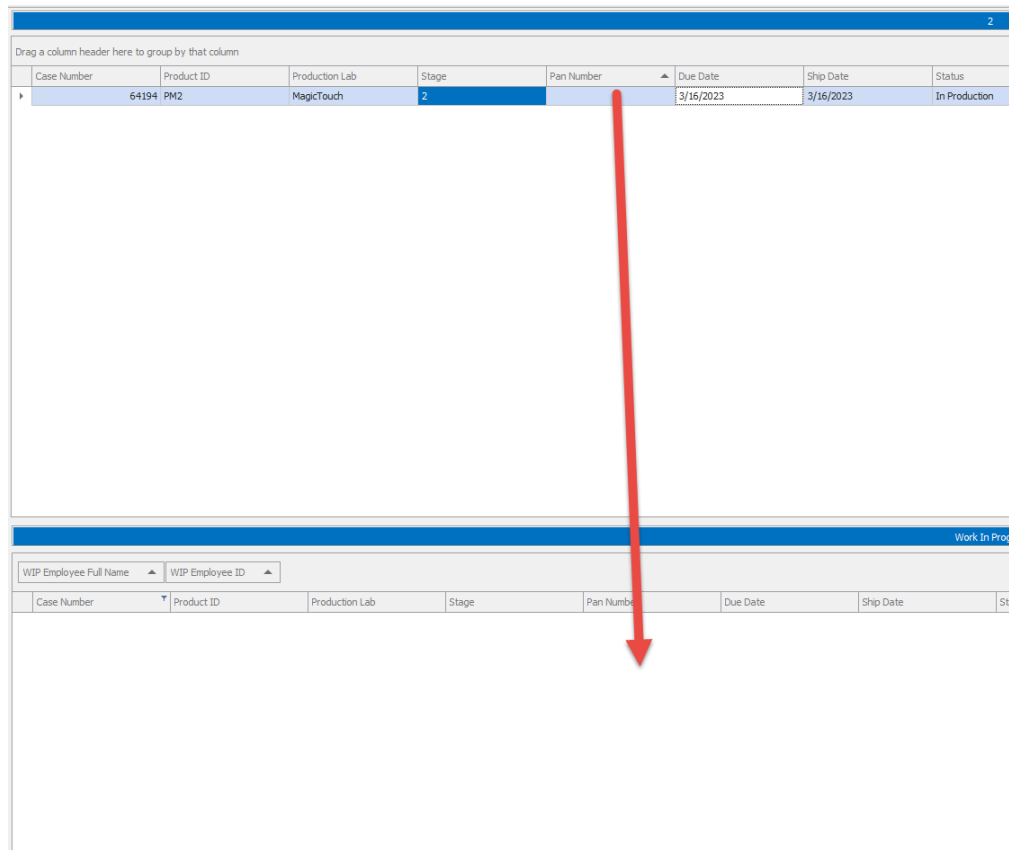
The 'Complete Task' dialog box is a light gray window with a title bar containing a green plus icon, the text 'Complete Task', and standard window controls (minimize, maximize, close). Inside the dialog, there are three rows of controls. The first row is labeled 'Technician' with a person icon and a dropdown menu showing 'Manuel Margot'. The second row is labeled 'Complete Date' with a calendar icon and a dropdown menu showing '03/02/2023'. The third row is labeled 'Complete Time' with a clock icon and a time selection control showing '09:37:09 AM'. At the bottom right, there are two buttons: 'Complete' with a floppy disk icon and 'Cancel' with a red 'X' icon.

6. Notes - Read notes that are displayed in the case's "Notes and Instructions" menu.
7. Write Notes - Add notes per stage. These will remain in Production Management and can be reviewed using the Write Notes feature as users write notes for each stage.
8. Attach Document
 - a. Allows attaching files to the cases. Users can upload files from their local machine, add a picture using a camera connected to the PC and add a URL.
 - b. Choose the department and related repository.
 - c. Choose to add visibility from the Customer Portal and add a Description.
 - d. Click Upload to send the files.



The 'Attach Document' dialog box is a light gray window with a title bar containing a green plus icon, the text 'Attach Document', and standard window controls (minimize, maximize, close). Inside the dialog, there are three buttons at the top: 'Upload a file' with a green up arrow icon, 'Camera' with a camera icon, and 'Add URL' with a chain link icon. Below these buttons are two dropdown menus: 'Department:' and 'Repository:' (showing 'Repository is not selected'). There is a checkbox labeled 'Allow Customer to View in Customer Portal' which is currently unchecked. Below the checkbox is a text area labeled 'Description'. At the bottom, there are two buttons: 'Upload' and 'Cancel'.

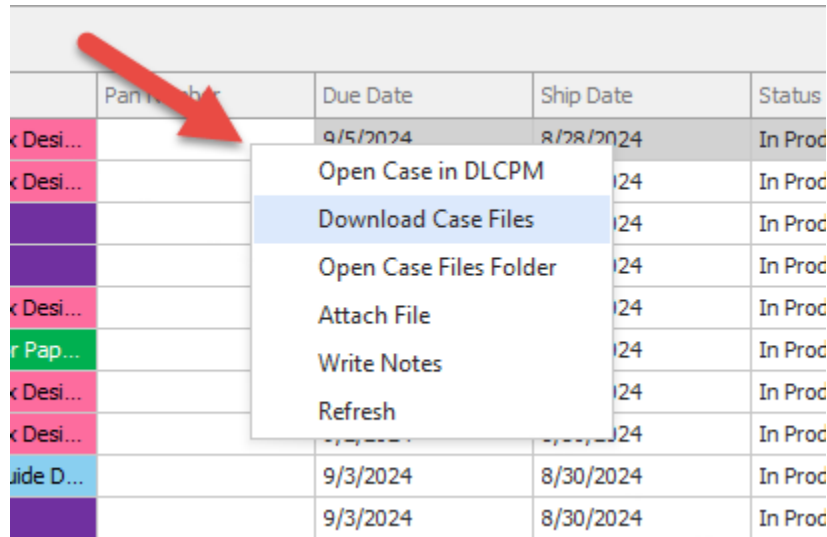
9. Move to WIP - Users can add groups of cases to the Work in Progress bar below the cases/product grid.



10. Complete Stage - All tasks from the current stage will be completed.
11. End WIP - This will move the case/products back into the main grid above.
12. Scan Cases - This option can be used to complete the case/product's current stage in bulk.

Using the Right click menu

Right-click on any case in the grid to bring up a list of case specific functions.



Part Number	Due Date	Ship Date	Status
Desi...	9/5/2024	8/28/2024	In Proc
Desi...		24	In Proc
		24	In Proc
		24	In Proc
Desi...		24	In Proc
Pap...		24	In Proc
Desi...		24	In Proc
Desi...		24	In Proc
ide D...	9/3/2024	8/30/2024	In Proc
	9/3/2024	8/30/2024	In Proc

1. Open Case in DLCPM – If the user is logged into DLCPM, this will open the case in question.
2. Download Case Files – Use this to download all related case files to your local machine.
3. Open Case Files Folder – This will open the related folder for the case. This may require access to the lab's server.
4. Attach File – Allows user to upload files to DLCPM from Production Management.
5. Write Notes – Case notes can be recorded by stage by using this option.
6. Refresh – Refreshes the grid.

Dashboard view

The production management Dashboard will give the user an overview of all production specific statistics for the current state of production.

To access the Dashboard, click the tab labeled Dashboard at the top of the menu. This will display a menu where you can choose a pre-defined query. Choose a query and click refresh.

The screenshot shows the 'Dashboard' tab selected. A dropdown menu is set to 'Completed Tasks', and a 'Refresh' button is visible. A yellow tooltip explains: 'Present a dashboard with the number of completed tasks for each technician, showcasing their performance for both today and the past week.'

Below the filters, a summary table displays completed tasks:

Task	3 Days Ago	Today	Grand Total
101045	1	7	8
101050		5	5
101062		2	2
101065	1	1	2
Grand Total	2	15	17

A red arrow points from the 'Today' column of the summary table to a detailed task list below. The list has a header: 'Drag a column header here to group by that column'.

Task	Description	Product ID	Case Number	Task Duration
101045	Rx Design CLDMO	1001	63610	
101045	Rx Design CLDMO	1001	66601	
101045	Rx Design CLDMO	1001	63609	
101045	Rx Design CLDMO	1001	66598	
101045	Rx Design CLDMO	1001	66600	
101045	Rx Design CLDMO	1001	66602	
101045	Rx Design CLDMO	1001	66599	

In the example above, the Dashboard displays all the completed tasks for the past week. Double-click a statistic and it will load in all the completed tasks.